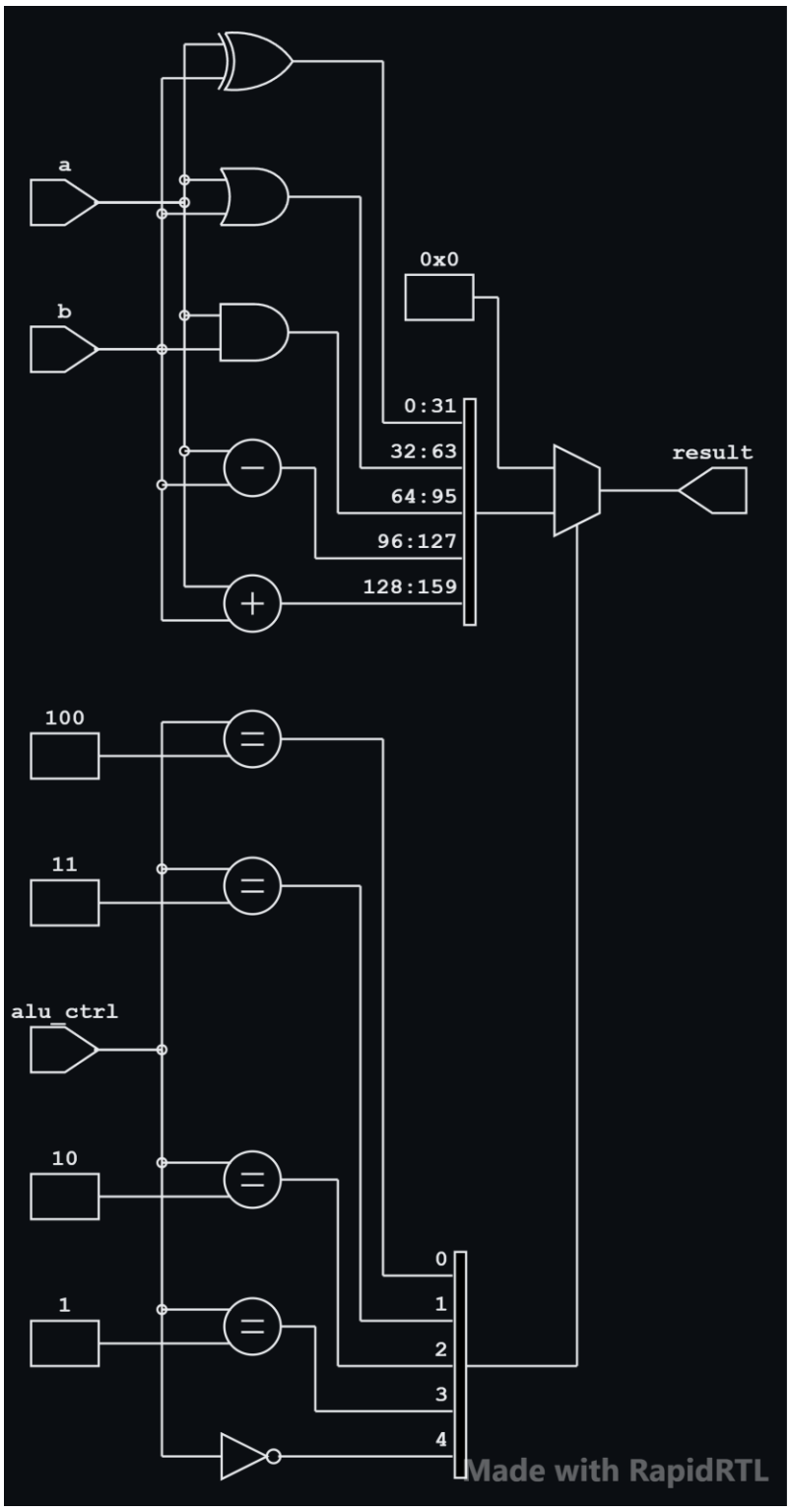
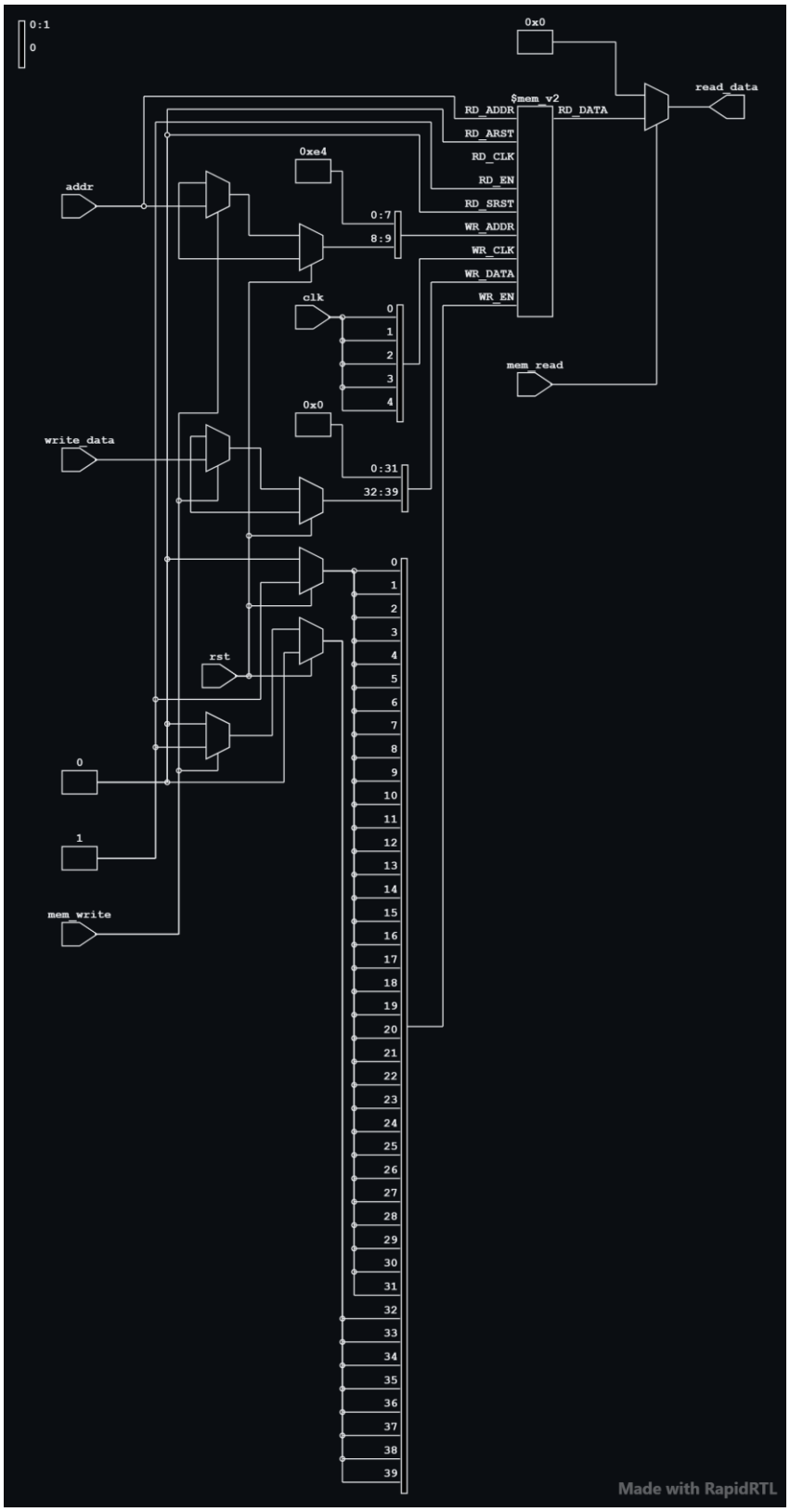
GATE LEVEL MODELING
using site https://www.rapidrtl.com/

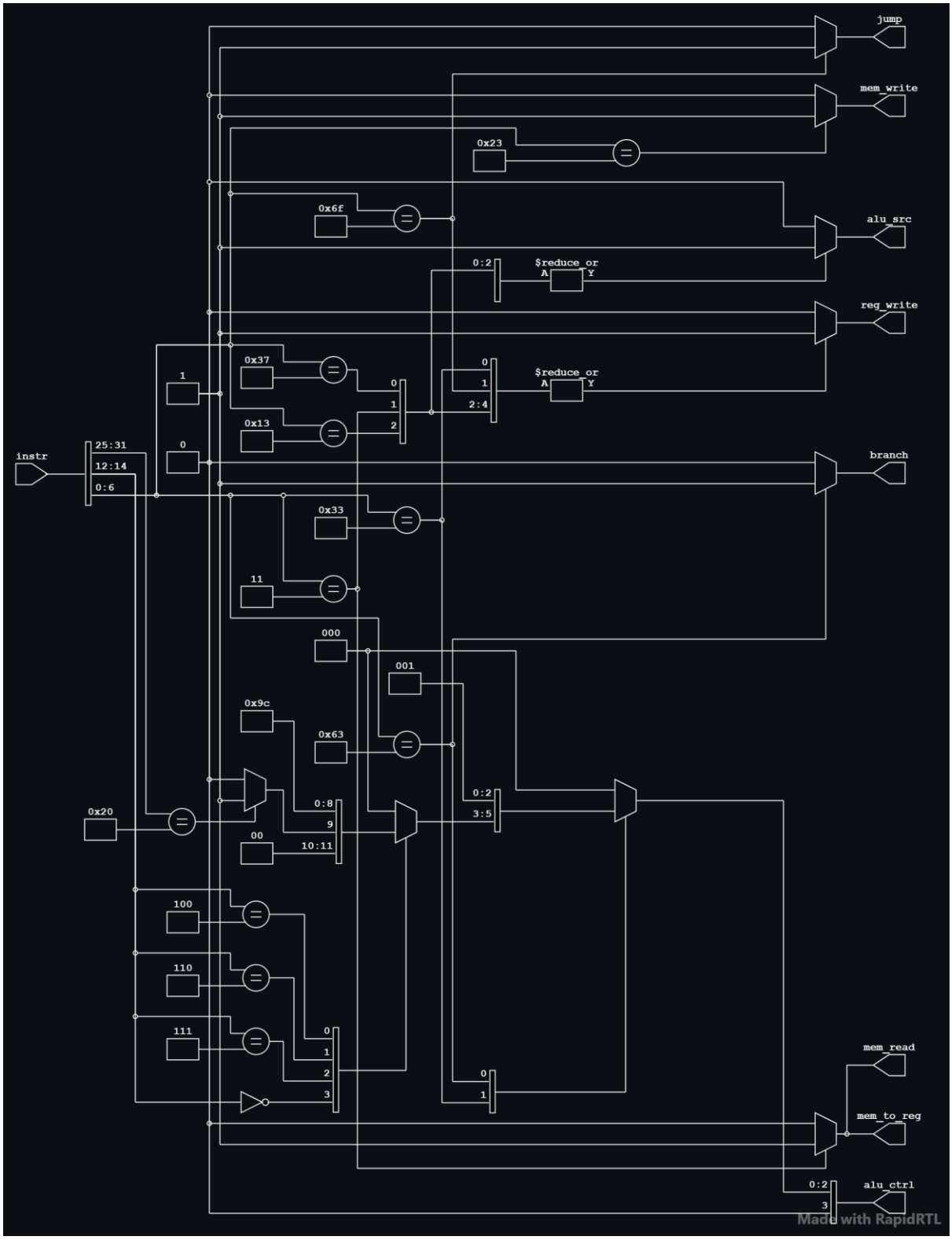
ALU unit



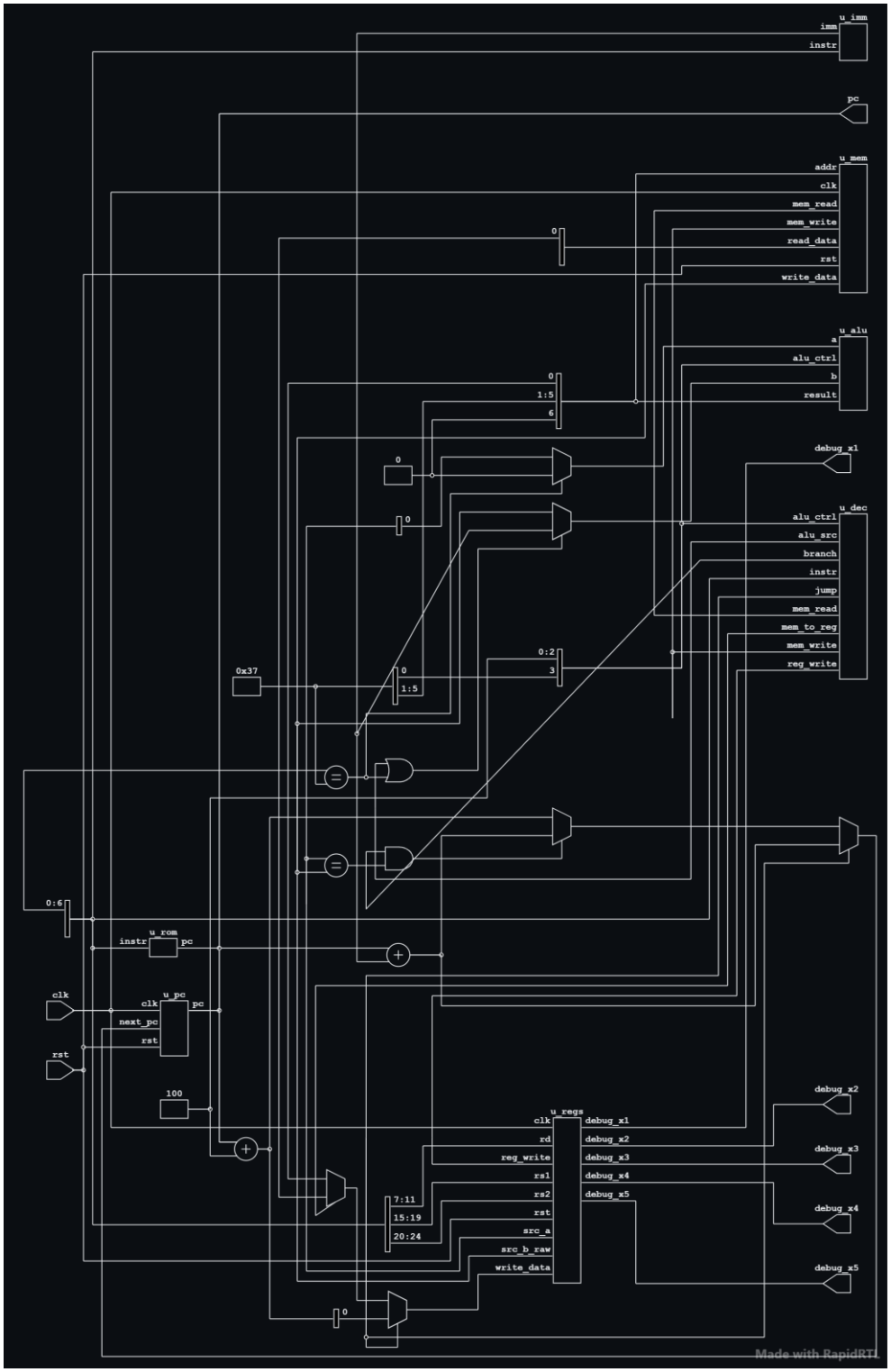
DATA memory


0:1
0
0x0
read_data
$mem_v2
RD_ADDR
RD_DATA
RD_ARST
RD_CLK
RD_EN
RD_SRST
WR_ADDR
WR_CLK
WR_DATA
WR_EN
0xe4
addr
0:7
8:9
clk
0
1
2
3
4
mem_read
0x0
write_data
0:31
32:39
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
rst
0
1
mem_write
Made with RapidRTL

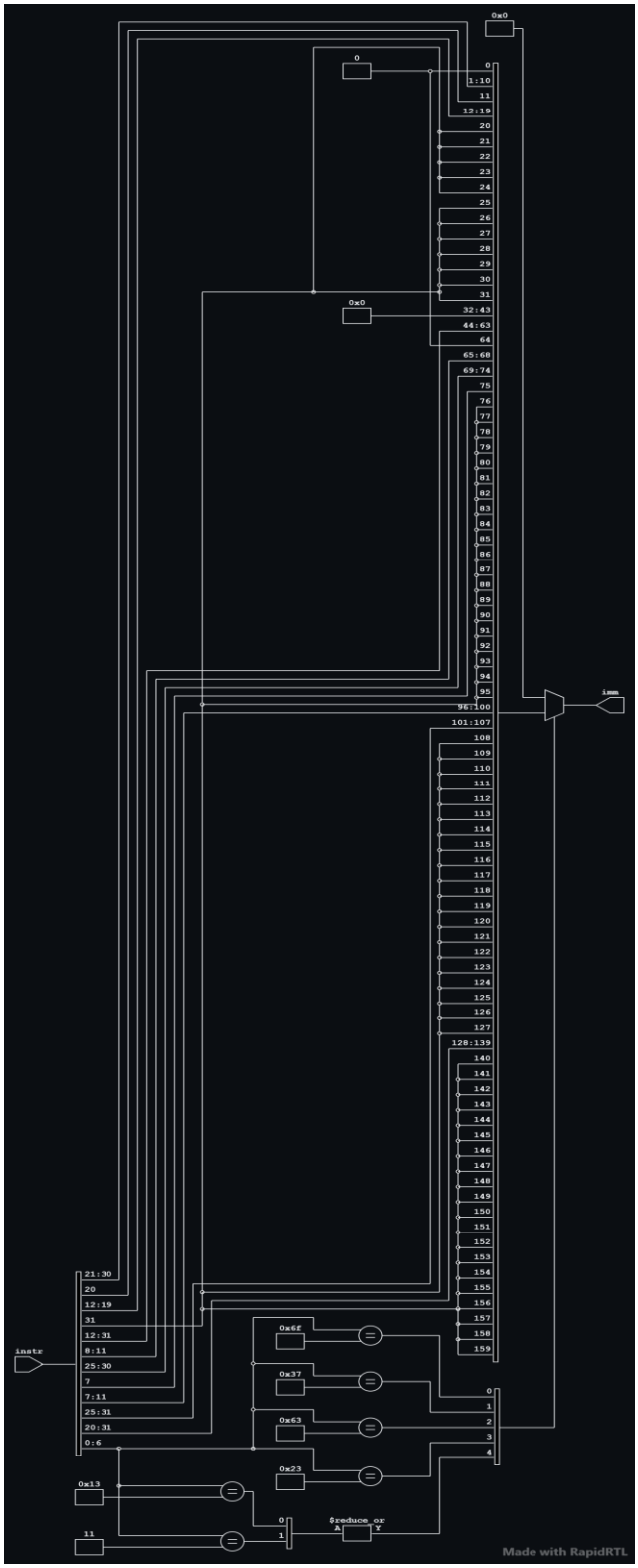
Decoder


jump
mem_write
0x23
0x6f
alu_src
0:2
$reduce_or
A
Y
reg_write
0x37
1
0
1
2
0
1
2:4
$reduce_or
A
Y
0x13
instr
25:31
12:14
0:6
0
branch
0x33
11
000
001
0x9c
0x63
0x20
0:8
9
00
10:11
0:2
3:5
100
110
111
0
1
2
3
0
1
mem_read
mem_to_reg
0:2
3
alu_ctrl
Made with RapidRTL

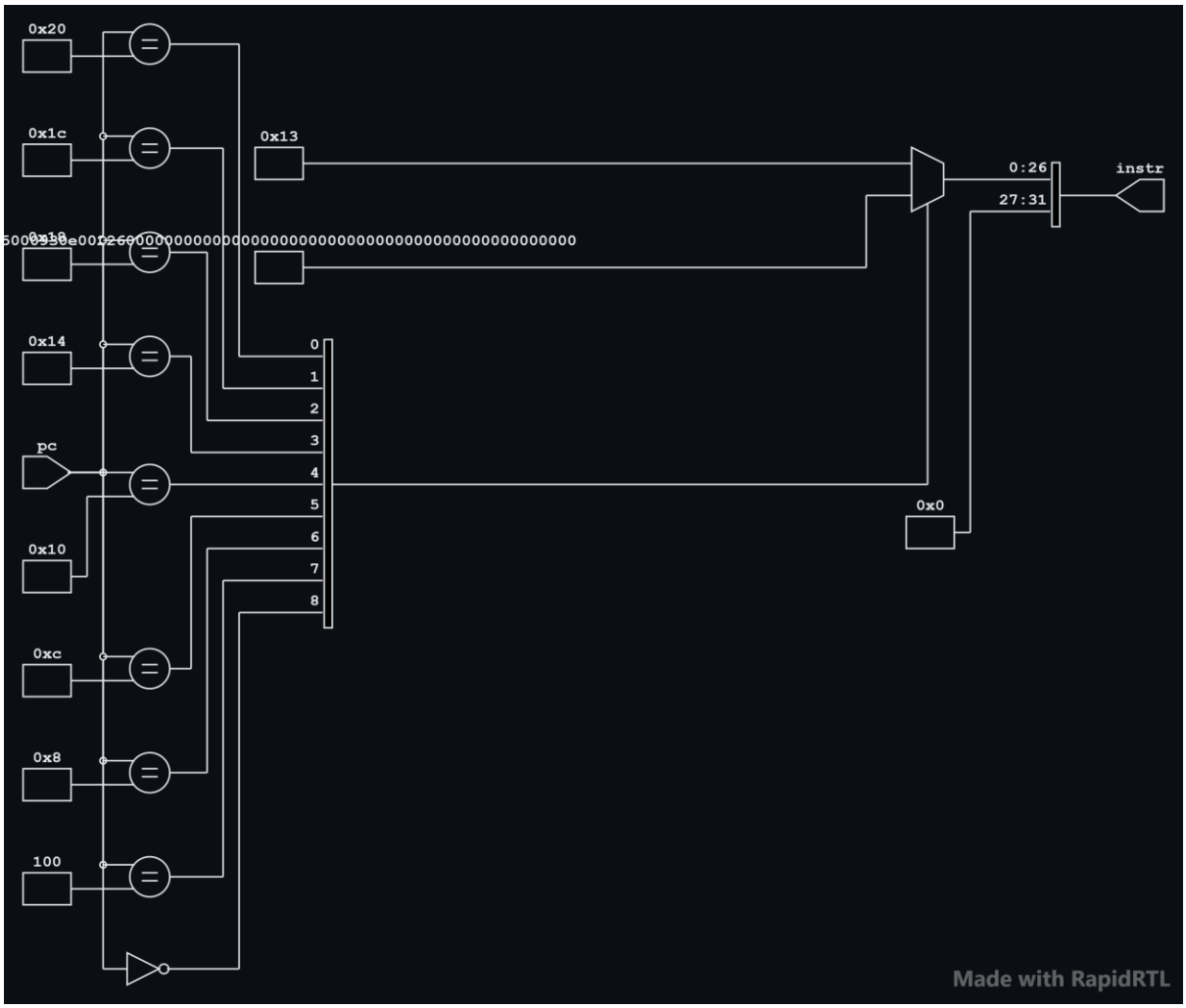
RISCV cpu


u_imm
imm
instr
pc
u_mem
addr
clk
mem_read
mem_write
read_data
rst
write_data
u_alu
a
alu_ctrl
b
result
0
1:5
6
debug_x1
u_dec
alu_ctrl
alu_src
branch
instr
jump
mem_read
mem_to_reg
mem_write
reg_write
0:2
3
0x37
0:6
u_rom
instr
pc
u_pc
clk
pc
next_pc
rst
100
u_regs
clk
rd
reg_write
rs1
rs2
rst
src_a
src_b_raw
write_data
debug_x1
debug_x2
debug_x3
debug_x4
debug_x5
7:11
15:19
20:24
Made with RapidRTL

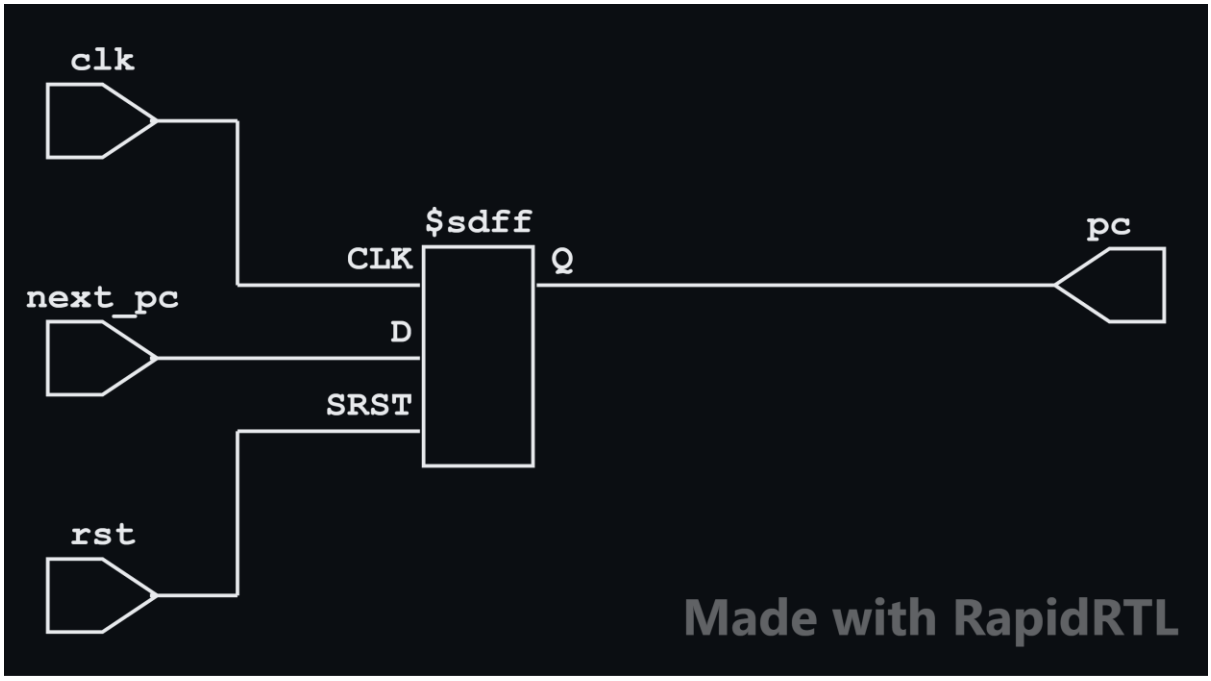
Immediate gen



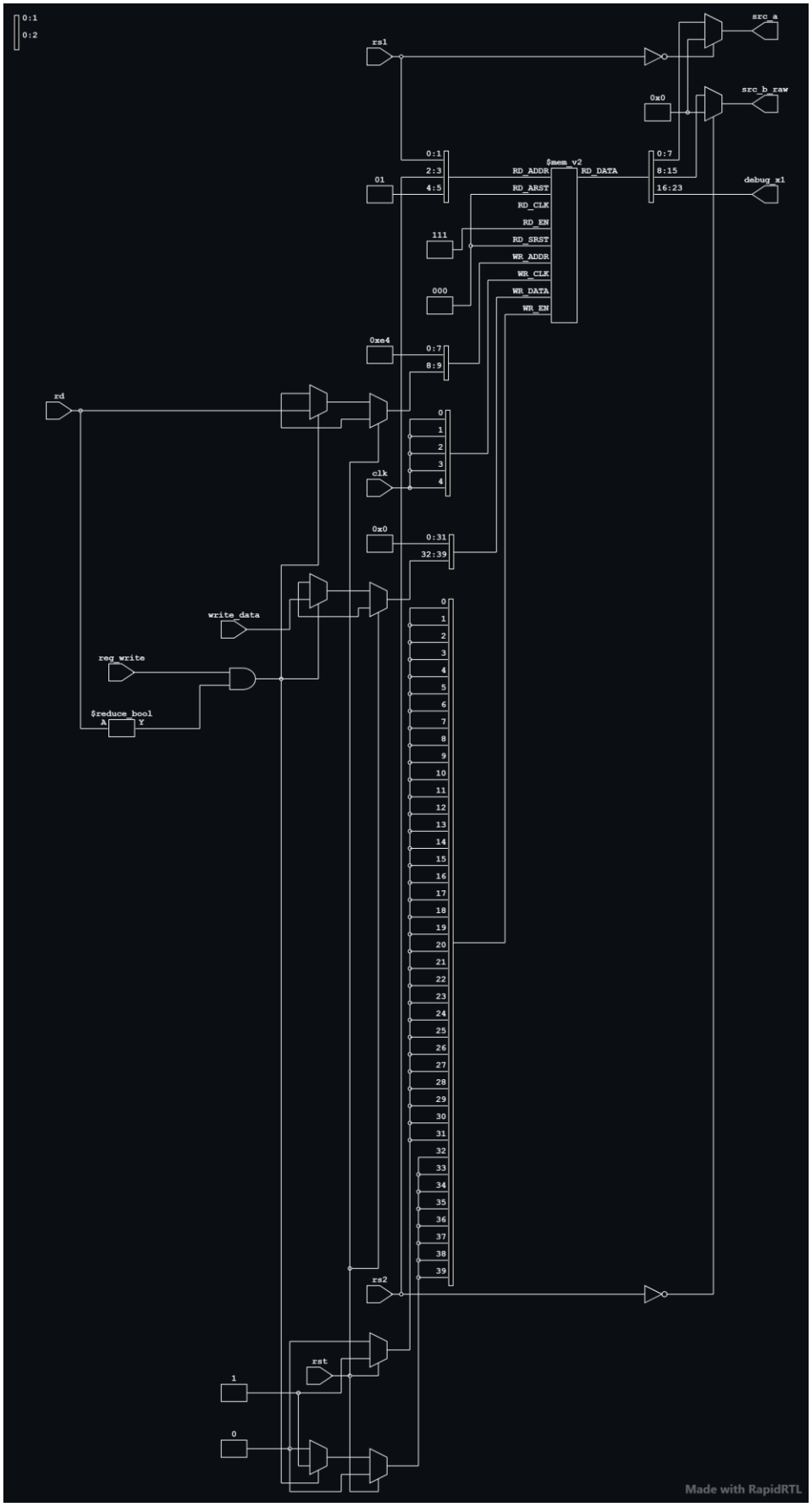
Instruction ROM



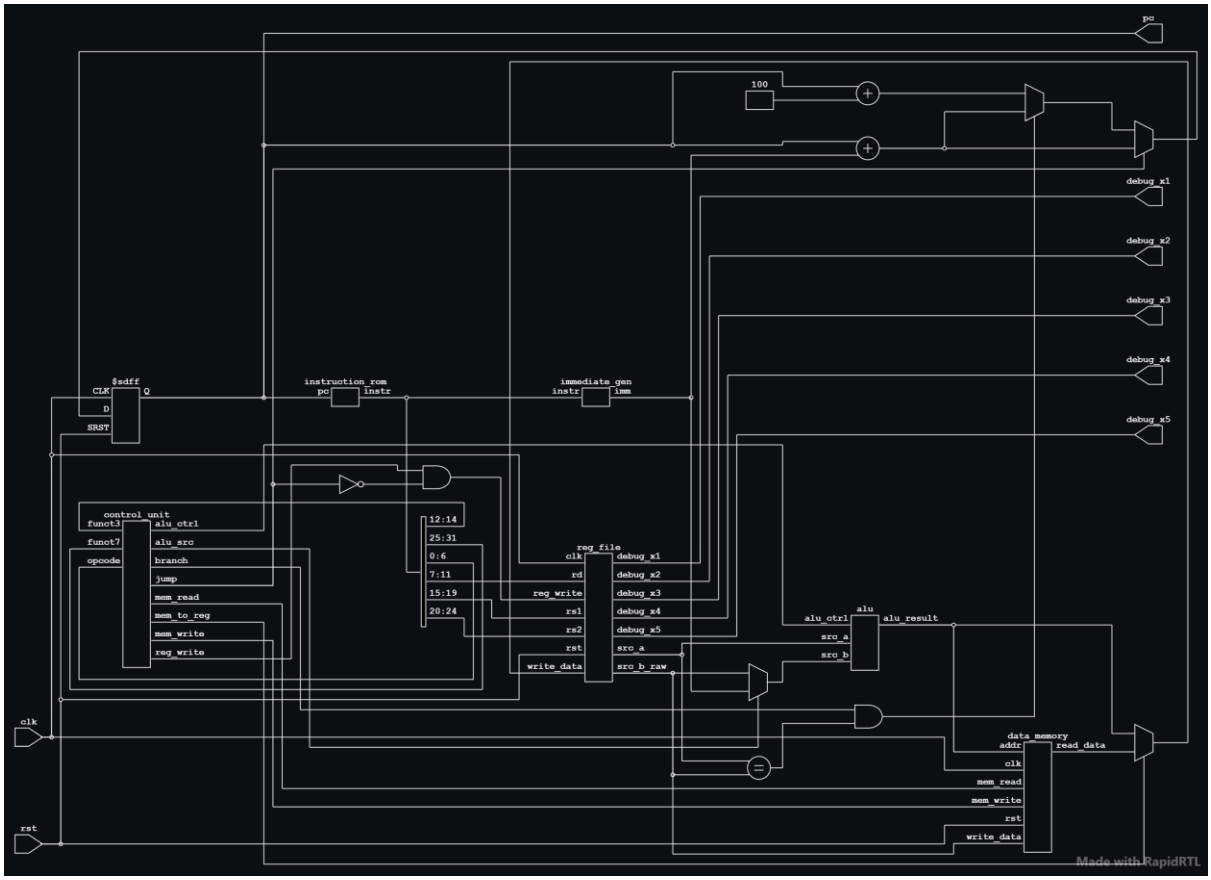
PROGRAM counter



REG FILE


0:1
0:2
rs1
src_a
0x0
src_b_raw
0:1
2:3
4:5
01
$mem_v2
RD_ADDR
RD_DATA
RD_ARST
RD_CLK
RD_EN
RD_SRST
WR_ADDR
WR_CLK
WR_DATA
WR_EN
0:7
8:15
16:23
debug_x1
111
000
0xe4
0:7
8:9
rd
0
1
2
3
4
clk
0x0
0:31
32:39
write_data
reg_write
$reduce_bool
A
Y
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
rs2
rst
1
0
Made with RapidRTL

COMPLETE


pc
100
debug_x1
debug_x2
debug_x3
debug_x4
debug_x5
$sdff
CLK
D
SRST
Q
instruction_rom
pc
instr
immediate_gen
instr
imm
control_unit
funct3
funct7
opcode
alu_ctrl
alu_src
branch
jump
mem_read
mem_to_reg
mem_write
reg_write
12:14
25:31
0:6
7:11
15:19
20:24
reg_file
clk
rd
reg_write
rs1
rs2
rst
write_data
debug_x1
debug_x2
debug_x3
debug_x4
debug_x5
src_a
src_b_raw
alu
alu_ctrl
src_a
src_b
alu_result
data_memory
addr
clk
mem_read
mem_write
rst
write_data
read_data
clk
rst
Made with RapidRTL